



College of Natural and Computational Sciences

Information Science School

Group Assignment

On

Neural Re-Ranking for Information Retrieval

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Date 4/26/2026

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Abstract

This project explores re-ranking methods in the context of information retrieval. Information Retrieval (IR) systems traditionally rely on lexical matching techniques such as BM25 to retrieve relevant documents. However, these methods struggle to capture semantic relationships between queries and documents. Neural re-ranking approaches address this limitation by applying deep learning models to reorder initially retrieved documents based on contextual understanding. This study investigates the effectiveness of neural re-ranking using a Cross-Encoder architecture implemented with Sentence Transformers with the MS MARCO dataset. The system first retrieves candidate passages using BM25 and then re-ranks them using a neural model. Performance is evaluated using Precision@10, Recall@10, Mean Reciprocal Rank (MRR@10), and Normalized Discounted Cumulative Gain (NDCG@10). Experimental results demonstrate that neural re-ranking significantly improves ranking quality compared to the BM25 baseline, particularly in terms of NDCG, Precision@10 and Recall@10 metrics. The experiment also shows that MRR@10 slightly decreases.

1. Introduction

Information Retrieval Systems (IRS) have dramatically changed the way how people acquire information for their need. Information Retrieval (IR) enables user to find relevant document from collection of different resources and widely used in search engines, digital libraries, and question answering systems.

Traditional IR approaches rely on lexical matching algorithms such as TF-IDF and BM25 to retrieve relevant documents. While these techniques are computationally efficient they often fail to capture semantic similarities between queries and documents. A critical component within this system is the re-ranking stage. Re-ranking carefully reorders these candidates to show the user the most relevant results after an initial, often rapid, retrieval phase that produces an extensive collection of candidate documents. This dramatically improves the overall quality and efficacy of the IR system.

Recent advances in deep learning have enabled neural models to learn contextual representations of text. Neural re-ranking has emerged as an effective approach where a neural network is used to re-order documents retrieved by a traditional search algorithm.

In this work, we explore neural re-ranking using a Cross-Encoder architecture implemented with Sentence Transformers. The MS MARCO dataset is used to train and evaluate the model. The objective of this study is to compare the performance of BM25 baseline retrieval with a neural re-ranking model.

2. Problem Statement

Traditional information retrieval systems primarily rely on traditional based ranking models such as BM25 and tf-idf which use lexical matching techniques to retrieve and rank documents. These models are efficient and widely adopted. They are limited in capturing semantic relationships between queries and documents. Traditional models like BM25 rely on exact keyword matching which fails if a query uses synonyms or different phrasing than the document.

In addition when relevant documents are retrieved their ranking positions may not reflect their true relevance and reducing the overall effectiveness of the retrieval system. This research addresses the problem of improving document ranking quality by applying neural re-ranking techniques specifically a Cross-Encoder model to reorder candidate documents retrieved by BM25. The goal is to enhance semantic understanding and ensure that relevant documents appear at higher ranks.

3. General Objective and Specific Objectives

General Objective

To improve the ranking quality of information retrieval systems by integrating neural re-ranking models with traditional BM25 retrieval.

Specific Objectives

- ❖ To implement a baseline retrieval system using BM25
- ❖ To design a two-stage retrieval pipeline
- ❖ To develop a neural re-ranking model using a Cross-Encoder model.
- ❖ To evaluate the performance of the proposed system using Precision@k, Recall@k, MRR@k, and NDCG@k.
- ❖ To compare BM25 and neural ranking approaches.

4. Reviewed Related work

Re-ranking has emerged as a promising approach to improve the quality of results in semantic search and recommendation systems. While early work in information retrieval focused primarily on traditional learning-to-rank methods, recent advances in transformer-based models have led to significant improvements in ranking accuracy. In this section, we review relevant work in three areas central to our project BM25, Deep Learning Techniques for Re-ranking and evaluation practices for re-ranking model.

4.1 Retrieval ranking model

Retrieval models take as input a user's query and then present a set of documents that are relevant to the query. In order to return a useful set of documents to the user, the retrieval model should be able to rank documents based on the given query.

Many neural ranking models have complex architectures, therefore computing the query-document relevance score using the neural ranking model for every document in the initial large collection of documents leads to a significant increase in the latency for obtaining a ranked list of documents from the user's side. So, the neural ranking component is usually used as a re-ranking step that takes two inputs which are the candidate documents and the processed query. The candidate documents are obtained from an unsupervised ranking stage, such as BM25 which takes as inputs the initial set of indexed documents and the processed query.

BM25 exact keyword matching is primarily used to rank a set of documents and the model works reasonably well as an initial ranker. Incorporating exact matching into neural ranking models can improve the retrieval performance mainly in terms of recall for keyword-based queries because as in traditional ad hoc document retrieval the document has more content than the query and the presence of query keywords in a document is an initial indicator of relevance. Both semantic and relevance matching signals are important to cover multiple scenarios of ad hoc retrieval tasks.

4.2 Deep Learning Techniques for Re-ranking

Deep learning models have significantly advanced the re-ranking task by capturing complex semantic relationships between queries and documents. BERT set new state-of-the-art performance on various sentence classification and sentence-pair regression tasks.

Transformer-based Re-rankers (BERT and T5) Transformer architectures dominate recent re-ranking advancements, especially BERT style encoders and T5-style sequence-to-sequence models.

BERT uses Cross-Encoders based and its variants jointly encode query-document pairs, enabling rich token-level interactions via self-attention and output a relevance score.

Cross-Encoder models jointly encode queries and documents, allowing them to capture rich semantic interactions. Unlike bi-encoder architectures that compute embedding's independently, Cross-Encoders evaluate the full query-document pair, making them particularly suitable for re-ranking tasks.

4.3 Evaluation practices for re-ranking model

Evaluation Metrics

Evaluation Metrics ranking quality is evaluated using Precision@k, Recall@k, MRR@k and NDCG@k metrics.

NDCG A metric that is commonly used to evaluate ranked results is Normalized Discounted Cumulative Gain (NDCG). It is a score between 0 and 1, where 1 is a perfect score.

MRR A third widely used metric for evaluating the quality of recommendation and information retrieval systems is Mean Reciprocal Rank (MRR). MRR is a simple metric and gives information about where on average the first relevant documents is in the ranked list. It is calculated as the Mean of Reciprocal Ranks (MRR) for all queries.

Precision is the ratio of correctly identified relevant items within the total recommended items inside the K-long list. Simply put, it shows how many recommended or retrieved items are genuinely relevant.

Recall measures the proportion of correctly identified relevant items in the top K recommendations out of the total number of relevant items in the dataset. In simpler terms, it indicates how many of the relevant items we could successfully find.

From our reviewed prior works showing the effectiveness of joint query document encoding in cross-encoders for improving ranking quality.

5. Methodology

In this section, we describe the datasets and models used in the experiments, as well as the training procedures applied to the models that required training. We also present the evaluation framework and explain the methodology used to ensure fair and consistent comparisons between retrieval approaches.

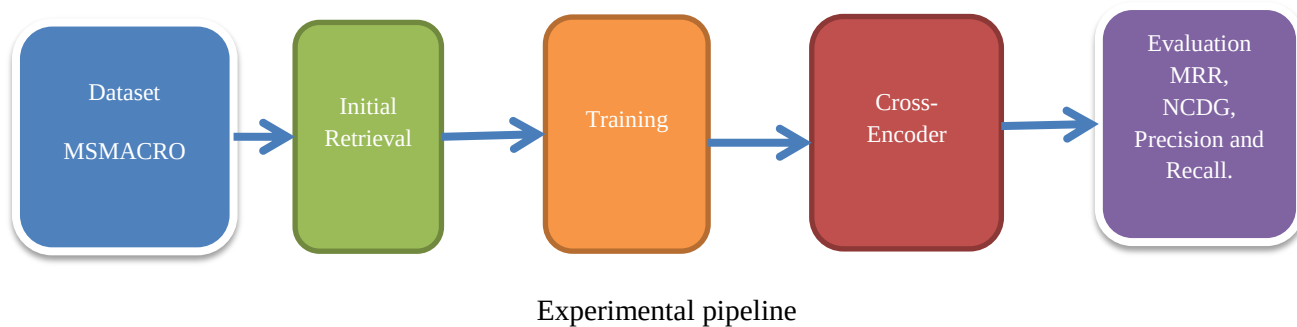
5.1 Research design

This study adopts **two-stage retrieval architecture** for passage ranking. In the first stage, a **BM25** lexical retrieval model is used to retrieve the top candidate passages for each query. In the second stage, a **neural re-ranking model** is trained to re-score the BM25 candidates and reorder them according to semantic relevance. This design is widely used in modern information retrieval because BM25 is efficient for initial retrieval, while neural re-rankers are effective when applied to a smaller candidate set rather than the entire corpus. Cross-encoder re-rankers are especially suitable for this purpose because they jointly process the query and passage and output relevance score for the pair.

This project follows **two-stage retrieval architecture** for passage ranking. In the first stage a **BM25** lexical retrieval model used as a baseline to retrieve the top candidate passages for each query. In the second stage a **neural re-ranking model** is trained to re-score the BM25 candidates and reorder them according to semantic relevance.

The objective of the methodology is to determine whether a neural re-ranking model can improve retrieval effectiveness over a BM25 baseline on the MS MARCO passage ranking task. The system is evaluated using standard ranking metrics including **MRR@10**, **Precision@10**, **Recall@10**, and **NDCG@10**.

The experimental Pipeline shows that from the starting to the end.



5.2 Dataset Collection

The experiments in this project utilize the **MS-MARCO passage ranking dataset** ("Microsoft / **MS-MARCO**", "v1.1") which is widely used for training and evaluating neural information retrieval systems. The dataset collection consists of real user queries collected from Bing search logs and corresponding passages extracted from web documents. This makes it suitable for evaluating both traditional lexical retrieval and neural ranking methods.

For this project, the dataset is organized into three core files

- ❖ **Passage collection:** contains `passage_id` and `passage_text`
- ❖ **Queries:** contains `query_id` and `query_text`
- ❖ **Relevance judgments:** contains `query_id`, `passage_id`, and relevance label

An example of dataset data that contain query and a passage marked as relevant, meaning it contains the answer to the query is and also there is non-relevant passage to the query.

Index	query	passage	label
0	what is rba	Since 2007, the RBA's outstanding reputation h...	0
1	what is rba	The Reserve Bank of Australia (RBA) came into ...	0
2	what is rba	RBA Recognized with the 2014 Microsoft US Regi...	0
3	what is rba	The inner workings of a rebuildable atomizer a...	0
4	what is rba	Results-Based Accountability® (also known as R...	0
5	what is rba	Results-Based Accountability® (also known as R...	1
6	what is rba	RBA uses a data-driven, decision-making proces...	0
7	what is rba	A rebuildable atomizer (RBA), often referred t...	0
8	what is rba	Get To Know Us. RBA is a digital and technolog...	0

5.3 Data preprocessing

For **BM25** lightweight text normalization is applied to both queries and passages. This includes lowercasing and punctuation cleaning. Tokenization breaking down passages and queries into individual words or tokens.

5.4 Dataset splitting

The dataset is split at the query level 70% training queries, 15% validation queries and 15% testing queries. The training set is used to learn model parameters, the validation set is used for hyper-parameter tuning and early stopping and the test set is used only for final evaluation.

The experiment Train (24,447), Validation (5238) and Test (5241) relevant and non-relevant passages.

5.5 BM25 baseline retrieval

BM25 is used as the baseline first-stage retriever. The BM25 index is built over the full passage corpus after tokenization. For each query BM25 calculates a lexical relevance score for every passage and returns the top-k highest-scoring candidates.

In this project BM25 serves two roles:

- ❖ Baseline retrieval system
- ❖ Candidate generator for the neural re-ranker

The BM25 implementation can be developed using a standard Python package such as rank-bm25, which provides Okapi BM25 and related variants for document ranking.

5.6 Preparing training data for neural re-ranking

The neural re-ranker requires both positive and negative training examples. Positive examples are obtained directly from the MSMARCO relevance judgments. However, non-relevant examples are not always explicitly available in sufficient quantity, so they must be generated.

In this study we use for each query two hard negative sampling. It is used to improve neural ranking performance.

- ❖ The relevant passage is used as a positive sample.
- ❖ Retrieved passages that are not labeled relevant are selected as negative samples.

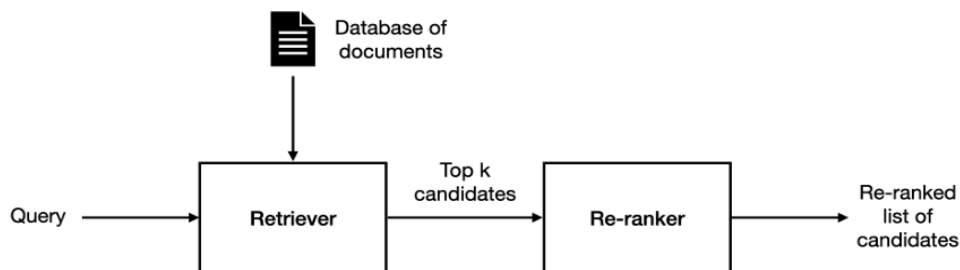
5.7 Neural Re-Ranking Model

The second stage of the retrieval is **BERT-based Cross-Encoders (neural re-ranker)**. It is common method for re-ranking using a cross-encoder. Cross-Encoder encodes a query together with each document retrieved from the initial retrieval and returns a score. It is neural Re-ranking and the neural re-ranking model is implemented using **Cross-Encoder architecture** from the Sentence Transformers library.

In this study, the re-ranker is trained using a **point wise ranking formulation**. Each query-passage pair is assigned a binary relevance label. During training, the model learns to produce high scores for relevant pairs and low scores for non-relevant pairs.

Cross-Encoder encodes a query together with each document retrieved from the initial retrieval and returns a score. It is neural Re-ranking and the neural re-ranking model is implemented using **Cross-Encoder architecture** from the Sentence Transformers library.

The output of the model is a single relevance score. During inference, all BM25 candidate passages for a query are scored by the model and then sorted in descending order to produce the final ranking.



A two stage ranking system where initial retrieval is done by the retriever and then re-ranked by there-ranker.

6. Training procedure for neural re-ranking

The training procedure consists of the following steps:

- ❖ Build the BM25 index from the passage collection.
- ❖ Retrieve BM25 top-k candidates for each training query.
- ❖ Generate positive and hard negative query-passage pairs.
- ❖ Convert the pairs into training, validation, and test sets.
- ❖ Train the cross-encoder model using labeled pairs.
- ❖ Monitor validation performance after each epoch.
- ❖ Save the best-performing checkpoint based on validation results.
- ❖ Evaluate after training

7. Evaluation Metrics

Ranking quality is evaluated using Precision@10, Recall@10, MRR@10, and NDCG@10 metrics. The cut-off at 10 was chosen because users typically focus on the top results when interacting with search systems, and we believe that this is a reasonable depth to expect users to look at. Moreover, using a standard cut-off allows for meaningful comparisons with previous studies in information retrieval. Limiting evaluation to the top 10 results also balances the trade-off between capturing user-relevant ranking quality and computational efficiency.

The system is evaluated using standard information retrieval ranking metrics:

- ❖ **MRR@10**: measures how early the first relevant result appears in the top 10.
- ❖ **Precision@10**: measures the proportion of relevant passages among the top 10 results.
- ❖ **Recall@10**: measures the proportion of relevant passages retrieved within the top 10.
- ❖ **NDCG@10**: measures ranking quality while rewarding relevant passages that appear higher in the ranked list

These metrics are computed for both:

- ❖ **BM25 baseline**
- ❖ **BM25 + neural re-ranking**

8. Summary of Results

The experimental results show that the BM25 + Cross-Encoder re-ranking model improves Precision@10, Recall@10, and NDCG@10 compared with the BM25 baseline. This indicates that the neural re-ranker improves the overall quality of the top-10 ranked results by placing more relevant passages within the retrieved set and improving their ranking positions. However, MRR@10 slightly decreases.

Overall, the Cross-Encoder provides a positive improvement in ranking quality, especially in Precision@10, Recall@10 and NDCG@10.

Model	Precision@10	Recall@10	MRR@10	NDCG@10
BM25	0.0722	0.7224	0.3377	0.429
BM25 + Cross-Encoder	0.0778	0.7785	0.3361	0.4411

Conclusion

From the Experiment results we concluded that combining BM25 with a neural Cross-Encoder re-ranker is more effective than using BM25 alone. BM25 remains a strong and efficient first-stage retriever while the neural re-ranker improves semantic relevance and ranking quality. Therefore, the proposed two-stage retrieval approach provides a practical and effective method for improving information retrieval performance.

Reference

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